Day 1

## I. Acid, Base and Salt

### A. Acid

- From the Latin word "Acidus" means "sour".
- Any substance that increases the concentration of the $H^+$ or hydrogen ion when it dissolves in water.

Examples:

- Hydrochloric acid (HCl)
- Sulphuric acid ($H_2SO_4$)
- Nitric acid ($HNO_3$)

### B. Base

- A base is any substance that increases the concentration of the $OH^-$ or hydroxide ion when it dissolves in water.

- Sodium hydroxide (NaOH)
- Potassium hydroxide (KOH)
- Calcium hydroxide ($Ca(OH)_2$)

### Acid and Base Taste

- An acid tastes sour, whereas a base tastes bitter.

The method of taste is not advised, as an acid or a base could be contaminated or corrosive.

**Example:** The flavors of curd, lemon juice, orange juice, and vinegar are all sour. Because they contain acids, these compounds have a sour flavor. Baking soda has a sour flavor. It's an example of a foundation.

### Properties of Acids and Bases

| Property | Acid | Base |
|---|---|---|
| **Taste** | sour | Bitter |
| **Feel** | Not slippery | Slippery, soapy |
| **Metals** | Mostly reacts with metals | Some reacts with metals |
| **Neutralization reaction** | Base to form salt | Acids to form salt |
| **ions** | Hydrogen (H+) | Hydroxide (OH-) |
| **pH scale** | 0 – 6.9 | 7.1 – 14 |
| **Litmus paper** | Turn blue litmus paper into red | Turn red litmus into blue |

**Acids and Bases**
An acid is any hydrogen-containing substance that is capable of donating a proton (hydrogen ion) to another substance. A base is a molecule or ion able to accept a hydrogen ion from an acid. Acidic substances are usually identified by their sour taste.

### Acid-Base Theory

- Pages 217-219

- Arrhenius Theory
- Bronsted-Lowry Theory
- Lewis Theory

**Activity 1:** Oral Activity

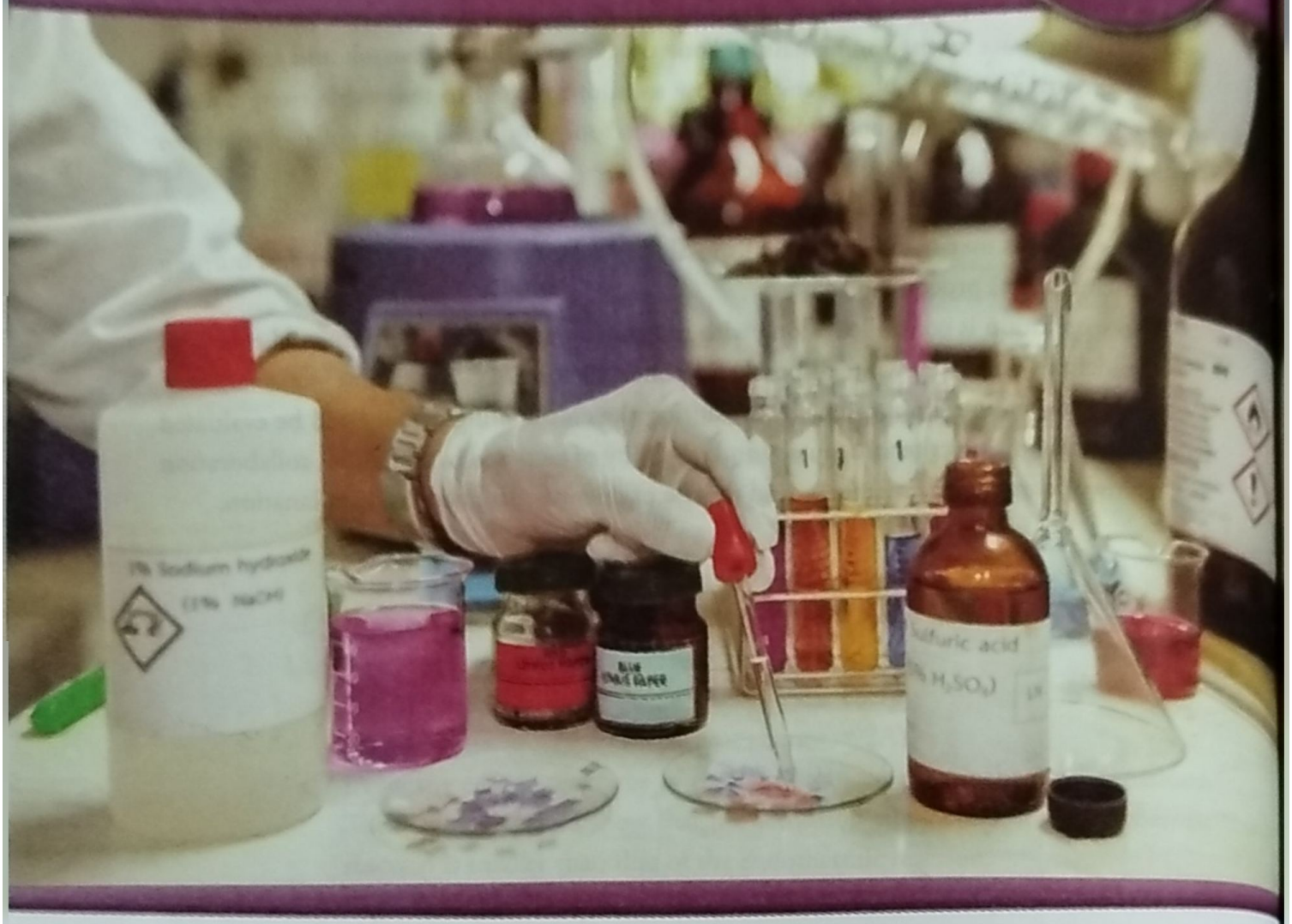
a. Determine whether the following are acid or base.

oral activity



# Acid-Base Reactions and Salts

CHAPTER 9



Whenever you combine an acid such as hydrochloric acid with a base like sodium bicarbonate, you are performing an acid-base reaction. This reaction forms salt and water through a process called *neutralization*, a type of double displacement reaction. Because acids and bases respond to indicators, these reactions can be easily verified, monitored, and controlled.

Acid-base reactions are significant in everyday activities. Some plants grow in acidic soil, but others do not. To neutralize acidic soil, calcium oxide (CaO) is used. Excess hydrochloric acid in gastric juice can cause acid reflux in the stomach. This condition is treated with antacids to neutralize excess HCl and avoid damage to the stomach. Sodium fluoride (NaF), a weak basic salt, is present in most toothpaste. Toothpaste helps prevent the growth of acidic bacteria that causes tooth decay.

Acids, bases, and salts play significant roles in everything—from digestion of the foods you eat to the function of the medicine you take, and even the cosmetic and cleaning products you use. Many of the food and household products in your home would not have much use without these substances. This chapter discusses acid-base reactions and the salts formed from this reaction, their significant role in biological, environmental processes, and applications in the food industry.

**LEARNING OBJECTIVES**

At the end of this chapter, you are expected to

1. describe acids and bases using the Arrhenius, Bronsted-Lowry, and Lewis theories;
2. explain the level of acidity and basicity of substances;
3. identify common acids and bases (e.g., hydrochloric acid, sodium hydroxide, and saline solution) using different indicators;
4. explain how acid-base reactions occur;
5. describe the formation of salts in acid-base reaction; and
6. identify the dependent and independent variables and control other variables in a valid and reliable scientific investigation.

# LESSON 1 The Acid-Base Theory

There are three different theories that describe and define acids and bases: the Arrhenius theory, the Bronsted-Lowry theory, and the Lewis theory.

## Arrhenius Theory

In the 19th century, Swedish scientist Svante August Arrhenius (1859–1927) was the first to describe acids and bases in terms of their behavior in water. According to Arrhenius:

$$\text{Arrhenius acid: } HCl_{(aq)} \rightarrow H^+_{(aq)} + Cl^-_{(aq)}$$

$$\text{Arrhenius base: } NaOH_{(aq)} \rightarrow Na^+_{(aq)} + OH^-_{(aq)}$$

Arrhenius acids are substances that produce hydrogen ions ($H^+$) when dissolved in water, while Arrhenius bases are ionic compounds that generate metal ions and hydroxide ions ($OH^-$) in water. The hydrogen ion ($H^+$) is responsible for the sour taste of acids, changes blue litmus paper to red, and corrodes some metals. The hydroxide ion ($OH^-$) gives most bases a bitter taste, turns red litmus paper blue, and gives a slippery feel. Table 9.1 compares some general properties of acids and bases.

**TABLE 9.1 GENERAL PROPERTIES OF ACIDS AND BASES**

| *Properties* | *Acids* | *Bases* |
|---|---|---|
| ions in water | $H^+$ (or $H_3O^+$) | $OH^-$ (Arrhenius) |
| behavior in water | $H^+$ donor | $H^+$ acceptor (Bronsted) |
| color in litmus | red | blue |
| color in phenolphthalein | colorless | violet |
| pH | below 7 | above 7 |
| neutralization | neutralizes bases | neutralizes acids |
| example | $HNO_3$, $H_2SO_4$, $HC_2H_3O_2$ | KOH, $Ca(OH)_2$ |

## Bronsted-Lowry Theory

In 1923, Danish chemist Johannes Nicolaus Bronsted (1879–1947), and English physical chemist Thomas Martin Lowry (1874–1936), expanded the definition of acids and bases beyond water solutions. A Bronsted-Lowry acid donates proton ($H^+$) to another substance, and a Bronsted-Lowry base accepts a proton. For example, the Bronsted-Lowry acid-base reaction is written as,

Reaction 1: $\underset{\text{acid}}{HCl_{(aq)}} + \underset{\text{base}}{H_2O_{(l)}} \rightarrow H_3O^+_{(aq)} + Cl^-_{(aq)}$

Reaction 2: $\underset{\text{base}}{NH_{3(aq)}} + \underset{\text{acid}}{H_2O_{(l)}} \rightarrow NH_4^+{}_{(aq)} + OH^-_{(aq)}$

In reaction 1, HCl donates a proton ($H^+$) to $H_2O$. In this case, $H_2O$ accepts a proton; thus, HCl acts as an acid while $H_2O$ acts as a base. In reaction 2, $NH_3$ accepts a proton ($H^+$) from $H_2O$; thus, $NH_3$ acts as a base while $H_2O$ as an acid.

## Lewis Theory

Also in 1923, American physical chemist Gilbert N. Lewis (1875–1946) proposed a broader theory to describe acids and bases. Lewis' theory explained the use of electron transfer which states that an acid accepts an electron pair while a base donates an electron pair.

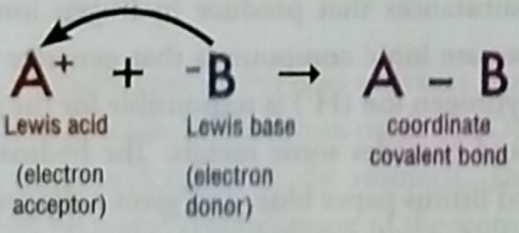



**Figure 9.1** Electron transfer showing Lewis acid-base theory, where a coordinate covalent bond forms when both electrons come from one atom.

A Lewis acid is described as "electron deficient" as it lacks a full electron pair and can accept one. It interacts with an "electron rich" substance called a Lewis base, which donates or shares its lone pair of electrons.

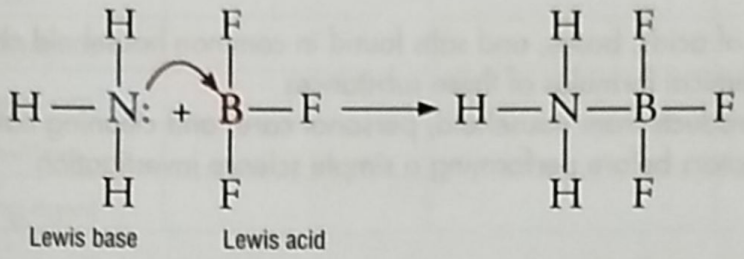



**Figure 9.2** The nitrogen atom in $NH_3$ provides both electrons to form the bond with boron in $BF_3$.

In this reaction shown in figure 9.2, $NH_3$ acts as a Lewis base by donating its lone pair of electrons, while $BF_3$ acts as a Lewis acid by accepting that pair. Table 9.2 is the summary of the description of acids and bases.

**TABLE 9.2 SUMMARIZED DEFINITIONS OF ACIDS AND BASES**

| *Properties* | *Acids* | *Bases* |
|---|---|---|
| Arrhenius | yields a proton ($H^+$) | yields a hydroxide ion ($OH^-$) |
| Bronsted-Lowry | donates a proton ($H^+$) | accepts a proton($H^+$) |
| Lewis | accepts an electron pair | donates an electron pair |

**FOLLOW-UP**

In each reaction, identify the acid and the base, and determine whether the reaction fits the Bronsted-Lowry theory or Lewis theory.

a. $NH_3 + H^+ \rightarrow NH_4^+$
b. $HBr + H_2O \rightarrow H_3O^+ + Br^-$
c. $CO_3^- + H_2O \rightarrow HCO_3^- + OH^-$
d. $OH^- + BF_3 \rightarrow HO^-BF_3$

Many food and household products contain a variety of chemical substances. To enhance the flavor of fruit juices, acids are often added. Vinegar, which contains about 4% acetic acid, is commonly used for pickling fruits. Antacids contain bases that help neutralize excess stomach acid. Salts are added to certain food products to extend shelf life and improve taste.

In the following activity, find out some of the substances listed in the nutrition facts and labels of some products available in your home.

## LESSON 2 Strengths of Acids and Bases

Acids and bases differ in the degree of their chemical activity in aqueous solutions. This difference is called *strength*, and it refers to how completely a substance separates into ions when dissolved in water. The strength of an acid or base affects its chemical behavior, especially in neutralization reaction.

### Strong and Weak Acids and Bases

Only a few acids are classified as strong acids, while most are weak acids. Strong acids such as HCl, $HNO_3$, and $H_2SO_4$ are strong electrolytes because they dissociate completely into ions in aqueous solutions. As discussed in acid-base theory, strong acids release all their hydrogen ions ($H^+$) when dissolved in water.

In contrast, weak acids only partially dissociate in solution and are therefore weak electrolytes. For example, acetic acid ($HC_2H_3O_2$), found in vinegar, does not fully separate into ions when dissolved in water.

Similarly, sodium hydroxide (NaOH) is a strong base and a strong electrolyte. It dissociates completely into sodium ions ($Na^+$) and hydroxide ions ($OH^-$) in solution. On the other hand, ammonia ($NH_3$) is a weak base because it does not fully dissociate into ions in water.

Strong Acid: $HNO_3 + H_2O \rightarrow H_3O^+ + NO_3^-$

Strong Base: $NaOH \rightarrow Na^+ + OH^-$

The strength of acids and bases varies. So how do you measure the degree of acidity or basicity (also called *alkalinity*) of substances in solution?

### The pH Scale

The pH scale is used to measure the acidity or basicity of a solution. It ranges from 0 to 14, with each value indicating the concentration of hydrogen ions ($H^+$) in the solution.

Acidic solution: $pH < 7$

Neutral solution: $pH = 7$

Basic solution: $pH > 7$

**TABLE 9.3 PH VALUES OF COMMON SUBSTANCES**

| | *pH* | *Example* |
|---|---|---|
| increasingly acidic ↑ | 0 | 1 M HCl |
| | 1 | gastric acid |
| | 2 | lemon juice |
| | 3 | apple juice |
| | 4 | tomato juice |
| | 5 | black coffee, banana |
| | 6 | milk |
| neutral | 7 | distilled water |
| increasingly alkaline ↓ | 8 | bile, egg |
| | 9 | baking soda |
| | 10 | milk of magnesia |
| | 11 | ammonia solution |
| | 12 | laundry soap, bleach |
| | 13 | oven cleaner |
| | 14 | 1 M NaOH |

In medical tests and laboratory experiments, a pH meter is often preferred for measuring acidity or basicity because it provides more precise and reliable readings than indicator papers. You can explore more interesting facts about pH meters in *Tech Talk* on page 229.

What happens to the pH of a solution after an acid-base reaction? The resulting pH depends on the relative strength of the acid and base involved. If a strong acid reacts with a strong base, the solution tends to become neutral. If one is stronger than the other, the resulting solution may still be slightly acidic or basic. Table 9.4 shows a glimpse of the pH outcomes from different acid-base reactions.

**TABLE 9.4 CHANGE IN PH AFTER AN ACID-BASE REACTION**

| *Type of acid-base reaction* | *Resulting pH* |
|---|---|
| strong acid + strong base | pH = 7, neutral |
| strong acid + weak base | pH < 7, slightly acidic |
| weak acid + strong base | pH > 7, slightly basic |
| weak acid + weak base | pH < 7 or pH > 7, either slightly acidic or basic |

## FOLLOW-UP

1. Predict and explain the change in pH after each acid-base reaction.
   a. $HCl$ (acid) + $NaOH$ (base) → $NaCl$ (salt) + $H_2O$ (water)
   b. $HC_2H_3O_2$ (acid) + $NH_3$ (base) → $NH_4C_2H_3O_2$ (salt) + $H_2O$ (water)
   c. $HNO_3$ (acid) + $NH_3$ (base) → $NH_4NO_3$ (salt) + $H_2O$ (water)
   d. $HC_2H_3O_2$ (acid) + $NaOH$ (base) → $NaC_2H_3O_2$ (salt) + $H_2O$ (water)
2. Teacher Emmy of grade 10-Rizal presented in class some household items with corresponding pH values. She then asked the class to do the following:
   a. Arrange in order of increasing acidity and explain the answers.

| *Household items* | *pH value* |
|---|---|
| shampoo | 8.0 |
| baking soda | 9.0 |
| bleach | 13.0 |
| hydrochloric acid | 1.0 |
| vinegar | 2.8 |
| carbonated beverage | 3.0 |

   b. Explain what is meant when "pH balanced" is written on the label of a shampoo bottle.

# LESSON 3 Salts as Product of Acid-Base Reactions

In nearly every kitchen, you will find a jar of salt. It is probably the world's oldest seasoning and food preservative, with a history almost as old as humankind itself. Salt was treasured by the ancient Greeks, Romans, and Egyptians not only for flavoring food but also for preserving it. Today, salt remains essential in daily life. According to industry reports, the global salt production reached over 270 million metric tons in 2022.

Because salt plays a vital role in cooking, food preservation, and processing, this lesson explores what salts are, how salts are formed, and common examples of salts and their general uses.

## Salts

*Salts* are ionic compounds formed when an acid reacts with a base, resulting in a neutralization reaction. Each salt is composed of positively charged ions (cations) and negatively charged ions (anions). These ions are held together by ionic bonds, forming a crystalline structure that makes the compound electrically neutral.

The most familiar example is sodium chloride (NaCl). In its solid form, NaCl appears as white crystals. When dissolved in water, these crystals dissolve and separate into sodium ($Na^+$) and chloride ($Cl^-$) ions, which allows its solution to conduct electricity.



(*a*) (*b*)

**Figure 9.3** (*a*) Sodium chloride has a crystalline structure and is commonly known as (*b*) table salt.

Since salts are ionic bonded, these compounds exhibit distinct physical and chemical properties. Table 9.5 shows some of the properties of salts.

**TABLE 9.5 SOME PROPERTIES OF SALTS**

| *Properties* | *Explanation* |
|---|---|
| brittleness | Salts are often brittle due to their rigid crystalline structure; they can shatter under pressure. |
| electrical conductivity | Salts conduct electricity when dissolved in water or in molten form, as ions can move freely. |
| ionic bonded | Salts are held together by strong electrostatic forces between oppositely charged ions. |
| melting and boiling points | High temperatures are needed to melt or boil salts due to the strong ionic bonds. |
| solubility | Most salts dissolve in water, separating into ions that interact with the solvent. |

## Acid-Base Reactions

Salts are formed from acid-base neutralization reactions. A salt is an ionic compound composed of a metal ion and a nonmetal or polyatomic ion. For example, when hydrochloric acid (HCl) reacts with sodium hydroxide (NaOH), the products are sodium chloride (NaCl) and water ($H_2O$):

$$HCl + NaOH \rightarrow NaCl + H_2O$$

To better understand the process of neutralization, you can examine its equation with the acid, base, and salt written in ionic form,

acid base salt water

$$H^+ + Cl^- + Na^+ + OH^- \rightarrow Na^+ + Cl^- + H_2O$$

In aqueous solution, HCl dissociates into $H^+$ (or $H_3O^+$) and $Cl^-$, and NaOH as $Na^+$ and $OH^-$. During neutralization, the $H^+$ (from the acid) reacts with $OH^-$ (from the base) to form water, $H_2O$, and the remaining $Na^+$ and $Cl^-$ ions stay in the solution. This reaction happens when a solution of any acid and a base, whether strong or weak, are completely neutralized. The pH of the solution becomes neutral.

## FOLLOW-UP

Complete and write the balanced equation for each acid-base neutralization reaction. Label the acids, bases, and salts.

1. $HNO_3 + KOH \rightarrow$ _____ $+ H_2O$
2. $HCl +$ ___ $\rightarrow CaCl_2 + H_2O$
3. $NaOH + H_3PO_4 \rightarrow$ ____ $+ H_2O$
4. $Ba(OH)_2 +$ ___ $\rightarrow BaCl_2 + H_2O$

Salts are essential ionic compounds found everywhere, from your kitchen to food and industrial plants. Some common salts and their uses are listed in table 9.6.

**TABLE 9.6 COMMON SALTS AND THEIR USES**

| *Salt* | *Use* |
|---|---|
| sodium chloride (NaCl) | for fluid balance and nerve function regulation in the body |
| sodium bromide (NaBr) | for chemical synthesis and water treatment |
| calcium chloride ($CaCl_2$) | as a dehumidifier or drying agent |
| potassium chloride (KCl) | in fertilizers as a source of potassium |
| potassium nitrate ($KNO_3$) | for curing meats and useful in the food industry<br>in fertilizers due to its high nitrogen |
| ammonium chloride ($NH_4Cl$) | source of nitrogen in fertilizers |
| magnesium sulfate ($MgSO_4$) | in bath salts to help relax muscles |
| sodium bicarbonate ($NaHCO_3$) | in fire extinguishers, puts out small fires by releasing carbon dioxide; acts as a leavening agent in baking |

There are some everyday neutralizations that you have to know that involve substances available at home. Taking antacids due to excess gastric acid in the stomach is one of them. Another is the removal of calcium carbonate deposits in kettles using acetic acid. Color change predicts the presence of acids, bases, and salts using phenolphthalein indicators. Do the following activity and enjoy the mystery behind the acid-base reaction.